

Supplementary Material for: Martingale Innovations from Contractive Recurrent Networks and Dimension-Robust Change-Point Detection

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A. Background and Preliminaries

We work throughout on a complete probability space $(\Omega, \mathcal{A}, \mathbb{P})$. All convergence statements are in the almost-sure or L^p sense unless otherwise noted.

A.1. Filtrations and the Information Flow

Definition A.1 (Filtration and natural filtration). A *filtration* is an increasing family of sub- σ -algebras $\mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \dots \subseteq \mathcal{A}$. For a sequence $(X_t)_{t \geq 1}$ of random variables, the *natural filtration* is $\mathcal{F}_t = \sigma(X_1, \dots, X_t)$ and its completion $\mathcal{F}_\infty^+ = \sigma(\bigcup_{t \geq 1} \mathcal{F}_t)$.

A *decreasing filtration* (or *reverse filtration*) is a family $\mathcal{G}_1 \supseteq \mathcal{G}_2 \supseteq \dots$ of sub- σ -algebras. The canonical example relevant to this paper is

$$\mathcal{G}_n = \sigma(X_n, X_{n+1}, \dots), \quad (\text{A.1})$$

which formalizes the information contained in the *tail* of the sequence starting at index n .

Definition A.2 (Tail σ -algebra). The *tail σ -algebra* of a sequence $(X_t)_{t \geq 1}$ is

$$\mathcal{T}_\infty = \bigcap_{n=1}^{\infty} \sigma(X_n, X_{n+1}, \dots) = \bigcap_{n=1}^{\infty} \mathcal{G}_n.$$

An event $A \in \mathcal{T}_\infty$ is *tail-measurable*: its occurrence or non-occurrence is not affected by any finite initial segment of the sequence.

Example A.3 (Tail-measurable functionals). For an i.i.d. or ergodic sequence (X_t) : (a) the limit $\lim_{n \rightarrow \infty} n^{-1} \sum_{t=1}^n X_t$ (when it exists) is $\mathcal{T}_\infty$ -measurable; (b) the event $\{\limsup_{t \rightarrow \infty} X_t > c\}$ is tail-measurable; (c) the population mean $\mu = \mathbb{E}[X_1]$ is a constant, hence trivially $\mathcal{T}_\infty$ -measurable under any probability measure. Finite-dimensional functions of $(X_1, \dots, X_k)$ for fixed k are *not* tail-measurable in general.

A.2. Martingales and Reverse Martingales

Definition A.4 (Martingale and martingale difference sequence). A sequence $(M_t, \mathcal{F}_t)_{t \geq 1}$ is a *martingale* if $\mathbb{E}[|M_t|] < \infty$ and $\mathbb{E}[M_{t+1} | \mathcal{F}_t] = M_t$ a.s. for all t . The sequence $(D_t)_{t \geq 1}$ is a *martingale difference sequence* (MDS) if $\mathbb{E}[|D_t|] < \infty$ and $\mathbb{E}[D_t | \mathcal{F}_{t-1}] = 0$ a.s. for all $t \geq 2$.

The posterior mean $\mathbb{E}[\theta | \mathcal{F}_t]$, for a fixed integrable parameter θ and observations $X_1, X_2, \dots$, is the canonical martingale in Bayesian sequential analysis.

Definition A.5 (Reverse martingale). Let $(\mathcal{G}_n)_{n \geq 1}$ be a decreasing filtration. A sequence $(Y_n, \mathcal{G}_n)_{n \geq 1}$ is a *reverse martingale* if $\mathbb{E}[|Y_n|] < \infty$ and

$$\mathbb{E}[Y_n | \mathcal{G}_{n+1}] = Y_{n+1} \quad \text{a.s. for all } n \geq 1.$$

A prototypical reverse martingale is obtained by taking an integrable random variable Y and setting $Y_n = \mathbb{E}[Y | \mathcal{G}_n]$ for a decreasing filtration $(\mathcal{G}_n)$. Exchangeability arguments also represent empirical averages through appropriate decreasing sigma-fields. The key convergence theorem is:

Theorem A.6 (Reverse martingale convergence; Doob 1953). If $(Y_n, \mathcal{G}_n)_{n \geq 1}$ is an L^1 -bounded reverse martingale with decreasing filtration $(\mathcal{G}_n)$, then

$$Y_n \xrightarrow{\text{a.s. and } L^1} Y_\infty := \mathbb{E}[Y_1 | \mathcal{G}_\infty], \quad \mathcal{G}_\infty = \bigcap_{n \geq 1} \mathcal{G}_n.$$

Proof. Let $(Y_n, \mathcal{G}_n)_{n \geq 1}$ be an L^1 -bounded reverse martingale with decreasing filtration $\mathcal{G}_1 \supseteq \mathcal{G}_2 \supseteq \dots$ and $\mathcal{G}_\infty = \bigcap_{n \geq 1} \mathcal{G}_n$. Without loss of generality, write $Y_n = \mathbb{E}[Y_1 | \mathcal{G}_n]$, which is consistent with the reverse martingale property by the tower law: for $m > n$, $\mathcal{G}_m \subseteq \mathcal{G}_n$ implies $\mathbb{E}[Y_n | \mathcal{G}_m] = \mathbb{E}[\mathbb{E}[Y_1 | \mathcal{G}_n] | \mathcal{G}_m] = \mathbb{E}[Y_1 | \mathcal{G}_m] = Y_m$ a.s.

Almost-sure convergence. For any $a < b$, let $U_n([a, b])$ be the number of upcrossings of $[a, b]$ by the sequence $Y_1, Y_2, \dots, Y_n$. Doob's upcrossing inequality for reverse martingales (Neveu, 1975, Chapter V) gives

$$(b - a) \mathbb{E}[U_n([a, b])] \leq \mathbb{E}[(Y_1 - a)^+] \leq \mathbb{E}[|Y_1|] + |a| < \infty.$$

Letting $n \rightarrow \infty$, the monotone convergence theorem gives $\mathbb{E}[U_\infty([a, b])] < \infty$, so $U_\infty([a, b]) < \infty$ a.s. Since $a < b$ are arbitrary rationals, the number of upcrossings of every rational interval is finite almost surely, which implies $Y_n \rightarrow Y_\infty$ a.s. for some $[-\infty, \infty]$ -valued random variable Y_∞ .

L^1 convergence and integrability of the limit. For each n , $|Y_n| = |\mathbb{E}[Y_1 | \mathcal{G}_n]| \leq \mathbb{E}[|Y_1| | \mathcal{G}_n]$ by Jensen's inequality. The family $\{\mathbb{E}[|Y_1| | \mathcal{G}_n]\}_{n \geq 1}$ is uniformly integrable: its L^1 norm is $\mathbb{E}[|Y_1|] < \infty$, and uniform integrability follows from the reverse martingale property of $(\mathbb{E}[|Y_1| | \mathcal{G}_n])$. Hence $\{Y_n\}$ is uniformly integrable. Uniform integrability together with a.s. convergence implies L^1 convergence: $\|Y_n - Y_\infty\|_{L^1} \rightarrow 0$, and in particular $Y_\infty \in L^1$.

Identification of the limit. For any $A \in \mathcal{G}_\infty \subseteq \mathcal{G}_n$, the definition of conditional expectation gives $\int_A Y_n d\mathbb{P} = \int_A Y_1 d\mathbb{P}$ for all n . Passing to the limit using L^1 convergence: $\int_A Y_\infty d\mathbb{P} = \int_A Y_1 d\mathbb{P}$ for all $A \in \mathcal{G}_\infty$. By the definition of conditional expectation, $Y_\infty = \mathbb{E}[Y_1 | \mathcal{G}_\infty]$ a.s. $\square$

Remark A.7 (Comparison with forward convergence). Doob's *forward* martingale convergence theorem requires L^1 -boundedness and yields almost-sure convergence but not necessarily L^1 convergence (unless the martingale is uniformly integrable). Reverse martingales are automatically uniformly integrable, so both modes of convergence are free. This property is useful for tail-projection arguments such as Lemma 2.1 of the main paper.

A.3. The Hewitt–Savage Zero-One Law

Theorem A.8 (Hewitt–Savage; Hewitt and Savage 1955). *Let $(X_t)_{t \geq 1}$ be an i.i.d. sequence. Then every event invariant under finite permutations of the coordinates has probability 0 or 1. In particular, every tail event $A \in \mathcal{T}_\infty$ satisfies $\mathbb{P}(A) \in \{0, 1\}$.*

For i.i.d. sequences, Theorem A.8 implies that the tail σ -algebra is trivial, so $\mathbb{E}[Z \mid \mathcal{T}_\infty] = \mathbb{E}[Z]$ for any integrable Z . The statement should not be extended to arbitrary exchangeable sequences: by de Finetti’s theorem, an exchangeable but non-i.i.d. Bernoulli sequence can have a nontrivial tail generated by the random mixing parameter. For stationary ergodic sequences, the ergodic theorem identifies limits of time averages, but it does not imply that every tail-measurable functional is literally a time average.

A.4. Recurrent Neural Networks and LSTMs

We give precise definitions of the two architectures whose hidden states we analyze.

Definition A.9 (Recurrent Neural Network). An RNN processes a sequence $(x_t)_{t=1}^T$, $x_t \in \mathbb{R}^p$, via

$$h_t = \phi(W_h h_{t-1} + W_x x_t + b_h), \quad (\text{A.2})$$

where $h_t \in \mathbb{R}^d$ is the *hidden state*, $W_h \in \mathbb{R}^{d \times d}$, $W_x \in \mathbb{R}^{d \times p}$, $b_h \in \mathbb{R}^d$, and $\phi : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is a coordinate-wise non-linear activation (e.g. tanh or ReLU). The initial state $h_0 \in \mathbb{R}^d$ is fixed or learnable; the L^2 theory below assumes it is square-integrable.

Assumption A.10 (Effective operator-norm contractivity). Throughout the paper we assume

$$L_\phi \|W_h\|_{\text{op}} < 1,$$

where L_ϕ is the Lipschitz constant of the activation ϕ and $\|W_h\|_{\text{op}} = \sigma_{\max}(W_h)$ is the operator norm. This is the contraction needed for L^2 -stability of (A.2). The spectral radius alone is insufficient for non-normal matrices because it does not control the vector norm. Spectral normalization can encourage or enforce this condition after accounting for L_ϕ (Miyato et al., 2018).

Definition A.11 (Long Short-Term Memory). An LSTM (Hochreiter and Schmidhuber, 1997) augments the basic RNN with a cell state $c_t \in \mathbb{R}^d$ and three gating mechanisms:

$$f_t = \sigma(W_f[h_{t-1}; x_t] + b_f), \quad (\text{A.3})$$

$$i_t = \sigma(W_i[h_{t-1}; x_t] + b_i), \quad (\text{A.4})$$

$$o_t = \sigma(W_o[h_{t-1}; x_t] + b_o), \quad (\text{A.5})$$

$$\tilde{c}_t = \tanh(W_c[h_{t-1}; x_t] + b_c), \quad (\text{A.6})$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t, \quad (\text{A.7})$$

$$h_t = o_t \odot \tanh(c_t), \quad (\text{A.8})$$

where $[h_{t-1}; x_t] \in \mathbb{R}^{d+p}$ denotes concatenation, σ is the logistic sigmoid, and $\odot$ is coordinate-wise multiplication. The *forget gate* $f_t \in (0, 1)^d$ controls how much of the previous cell state survives to time t .

Remark A.12 (Contractivity of LSTM). The coordinate-wise bound $f_t \in (0, 1)^d$ does not by itself make the full LSTM cell recursion uniformly contractive. The gate can be arbitrarily close to one, and f_t depends on h_{t-1} , which depends on c_{t-1} , so derivatives of the gate map also enter the Lipschitz constant. A rigorous contraction theorem for LSTMs therefore requires a separate assumption, such as a uniform gate bound $\|f_t\|_\infty \leq \gamma < 1$ together with Lipschitz control of the gate maps. In this paper, the rigorous contraction theorem is stated only for the basic RNN recurrence.

A.5. Stationary Ergodic Processes

Definition A.13 (Stationarity and ergodicity). A process $(X_t)_{t \geq 1}$ is (strictly) stationary if

$$(X_{t+k})_{t \geq 1} \stackrel{d}{=} (X_t)_{t \geq 1}$$

for all $k \geq 1$. It is *ergodic* if every shift-invariant event has probability 0 or 1. By the ergodic theorem (Walters, 1982), for any L^1 functional g ,

$$\frac{1}{T} \sum_{t=1}^T g(X_t) \xrightarrow{a.s.} \mathbb{E}[g(X_1)] \quad \text{as } T \rightarrow \infty.$$

Stationarity is the maintained assumption throughout Sections 2 and 4 of the main paper. The non-stationary extension via locally stationary processes (Dahlhaus, 1997) is treated in Section 2.3 of the main paper.

B. Proofs

B.1. Proof of Lemma 2.1 of the main paper (Equivalence of Forward and Tail Limits)

Proof of Lemma 2.1 in the main paper. Part (a). Since $\theta \in L^\infty \subseteq L^1$ and θ is $\mathcal{T}_\infty$ -measurable, it is also $\mathcal{F}_\infty^+$ -measurable because $\mathcal{T}_\infty \subseteq \mathcal{G}_1 = \sigma(X_1, X_2, \dots) \subseteq \mathcal{F}_\infty^+$. By Doob's L^1 martingale convergence theorem (Doob, 1953), $\mathbb{E}[\theta | \mathcal{F}_t] \rightarrow \mathbb{E}[\theta | \mathcal{F}_\infty^+]$ a.s. and in L^1 . Since θ is $\mathcal{F}_\infty^+$ -measurable, $\mathbb{E}[\theta | \mathcal{F}_\infty^+] = \theta$ a.s.

Part (b). The sequence $(\mathbb{E}[\theta | \mathcal{G}_n], \mathcal{G}_n)_{n \geq 1}$ is a reverse martingale: for $m > n$, $\mathcal{G}_m \subseteq \mathcal{G}_n$ implies $\mathbb{E}[\mathbb{E}[\theta | \mathcal{G}_n] | \mathcal{G}_m] = \mathbb{E}[\theta | \mathcal{G}_m]$ by the tower property. Boundedness of θ gives L^1 -boundedness. By the reverse martingale convergence theorem (Theorem A.6; Doob 1953), $\mathbb{E}[\theta | \mathcal{G}_n] \rightarrow \mathbb{E}[\theta | \mathcal{T}_\infty]$ a.s. and in L^1 . Since θ is $\mathcal{T}_\infty$ -measurable, $\mathbb{E}[\theta | \mathcal{T}_\infty] = \theta$ a.s.

Part (c). Immediate from (a) and (b): both limits equal θ a.s.

Parts (d) and (e). For general $\theta \in L^1$, part (a) extends to $\mathbb{E}[\theta | \mathcal{F}_t] \rightarrow \mathbb{E}[\theta | \mathcal{F}_\infty^+]$ by the same Doob argument. For part (e), the reverse martingale convergence in (b) applies unchanged since the reverse martingale is $(\mathbb{E}[\theta | \mathcal{G}_n], \mathcal{G}_n)$, and the limit is $\mathbb{E}[\theta | \mathcal{T}_\infty]$ regardless of whether θ is tail-measurable. The two limits agree if and only if $\mathbb{E}[\theta | \mathcal{F}_\infty^+]$ is $\mathcal{T}_\infty$ -measurable, which holds when θ itself is tail-measurable. $\square$

B.2. Proof of Theorem 2.5 of the main paper (MDS Decomposition with Stabilization Rate)

Proof of Theorem 2.5 in the main paper. Throughout, $\rho = L_\phi \|W_h\|_{\text{op}} < 1$ and $\Delta_0 = \|h_0 - H_0^*\|_{L^2}$.

Part (i): L^2 -boundedness. The framework assumes $\phi(0) = 0$ (as holds for tanh), so the Lipschitz bound gives $\|\phi(u)\| \leq L_\phi \|u\|$. (For activations with $\phi(0) \neq 0$, e.g. sigmoid, an additional term $\|\phi(0)\|/(1-\rho)$ appears in C_0 , but finiteness is unchanged; the stated C_0 formula covers the $\phi(0) = 0$ case.) By the Lipschitz bound $\|\phi(u)\| \leq L_\phi \|u\|$ and Minkowski's inequality in L^2 ,

$$\|h_t\|_{L^2} \leq L_\phi \|W_h h_{t-1} + W_x X_t + b_h\|_{L^2} \leq \rho \|h_{t-1}\|_{L^2} + L_\phi \|W_x\|_{\text{op}} \|X_t\|_{L^2} + L_\phi \|b_h\|.$$

Let $c := L_\phi (\|W_x\|_{\text{op}} \|X_1\|_{L^2} + \|b_h\|)$, which is finite by stationarity. Iterating the inequality and summing the geometric series gives

$$\|h_t\|_{L^2} \leq \rho^t \|h_0\|_{L^2} + \frac{c}{1-\rho},$$

so $\sup_t \|h_t\|_{L^2} \leq \|h_0\|_{L^2} + c/(1-\rho) =: C_0 < \infty$.

Part (ii): MDS structure. Since h_t is $\mathcal{F}_t$ -measurable and square-integrable by part (i), the conditional expectation $D_t := \mathbb{E}[h_t | \mathcal{F}_{t-1}]$ is well-defined and $\mathcal{F}_{t-1}$ -measurable. Setting $M_t := h_t - D_t$,

$$\mathbb{E}[M_t | \mathcal{F}_{t-1}] = \mathbb{E}[h_t - D_t | \mathcal{F}_{t-1}] = D_t - D_t = 0 \quad \text{a.s.}$$

Part (iii): Innovation L^2 bound. Conditional on $\mathcal{F}_{t-1}$, the previous state h_{t-1} is fixed and the only new random input is X_t . Put $\mu_t = \mathbb{E}[X_t | \mathcal{F}_{t-1}]$ and define $g_t(x) := \phi(W_h h_{t-1} + W_x x + b_h)$. Since conditional expectation is the L^2 projection onto $\mathcal{F}_{t-1}$,

$$\begin{aligned} \mathbb{E}\{\|M_t\|^2 | \mathcal{F}_{t-1}\} &\leq \mathbb{E}\{\|g_t(X_t) - g_t(\mu_t)\|^2 | \mathcal{F}_{t-1}\} \\ &\leq L_\phi^2 \|W_x\|_{\text{op}}^2 \mathbb{E}\{\|X_t - \mu_t\|^2 | \mathcal{F}_{t-1}\}. \end{aligned}$$

Taking unconditional expectations: since $\mathbb{E}\{\|X_t - \mu_t\|^2 | \mathcal{F}_{t-1}\} = \text{Var}(X_t | \mathcal{F}_{t-1})$ is the conditional variance, the law of total variance gives $\mathbb{E}[\text{Var}(X_t | \mathcal{F}_{t-1})] = \text{Var}_2(X_t) - \mathbb{E}\|\mu_t - \mathbb{E}X_t\|^2 \leq \text{Var}_2(X_t)$, which yields equation (1) in the main paper.

Part (iv): Innovation stabilization rate. Write $M_t - M_t^* = (h_t - H_t^*) - (D_t - D_t^*)$. By the triangle inequality, $\|M_t - M_t^*\|_{L^2} \leq \|h_t - H_t^*\|_{L^2} + \|D_t - D_t^*\|_{L^2}$. The coupling bound (Proposition B.1 below) gives $\|h_t - H_t^*\|_{L^2} \leq \rho^t \Delta_0$. Since conditional expectation is an L^2 contraction, $\|D_t - D_t^*\|_{L^2} = \|\mathbb{E}[h_t - H_t^* | \mathcal{F}_{t-1}]\|_{L^2} \leq \|h_t - H_t^*\|_{L^2} \leq \rho^t \Delta_0$. Adding the two bounds gives $\|M_t - M_t^*\|_{L^2} \leq 2\rho^t \Delta_0$.

Part (v): Variance decomposition and Σ_M^* . The law of total variance applied coordinatewise, combined with $\mathbb{E}[M_t | \mathcal{F}_{t-1}] = 0$, gives

$$\mathbb{E}\|h_t - \mathbb{E}h_t\|^2 = \mathbb{E}\|D_t - \mathbb{E}h_t\|^2 + \mathbb{E}\|M_t\|^2,$$

since the cross term $2\mathbb{E}\langle D_t - \mathbb{E}h_t, M_t \rangle = 0$.

For the stationary regime: stationarity of (H_t^*) and (X_t) implies stationarity of $(M_t^* = H_t^* - D_t^*)$, so $\Sigma_M^* := \mathbb{E}[M_t^* (M_t^*)^\top]$ is independent of t . Positive semi-definiteness is immediate. The bound $\Sigma_M^* \preceq \sigma_M^2 I_p$ follows from part (iii) applied to the stationary solution.

For strict positivity of Σ_M^* : suppose for contradiction that $\lambda_{\min}(\Sigma_M^*) = 0$. Then there exists a unit vector $v \in \mathbb{R}^p$ with $\mathbb{E}[(v^\top M_t^*)^2] = 0$, so $v^\top M_t^* = 0$ a.s., meaning $v^\top H_t^*$ is $\mathcal{F}_{t-1}$ -measurable. But then $\text{Var}(v^\top H_t^* | \mathcal{F}_{t-1}) = 0$ a.s., which contradicts the nondegeneracy condition (5) in the main paper.

(equation (5): $\text{Var}(v^\top \phi(W_h H_{t-1}^* + W_x X_t + b_h) \mid \mathcal{F}_{t-1}) > 0$ a.s. for all nonzero v). Hence $\lambda_{\min}(\Sigma_M^*) > 0$ under that condition.

Remark on the full row rank sufficient condition. A classical sufficient condition for equation (5) is $\text{rank}(W_x) = p$ together with X_t not entirely predictable from $\mathcal{F}_{t-1}$. When W_x has full row rank, $W_x^\top v \neq 0$ for all nonzero v , which implies $\text{Var}(v^\top H_t^* \mid \mathcal{F}_{t-1}) > 0$ a.s. for all nonzero v . This argument can fail when ϕ is a saturating nonlinearity in a nearly flat region; the nondegeneracy condition (5) is the correct sufficient condition in general. $\square$

B.3. Proof of Corollary 2.8 of the main paper (Functional CLT for MDS Innovations)

Proof of Corollary 2.8 in the main paper. CLT for the stationary process (M_t^*) . Under the stationary law, $(M_t^*, \mathcal{F}_t)_{t \geq 1}$ is a strictly stationary MDS. We verify $\sup_t \mathbb{E} \|M_t^*\|^{2+\delta} < \infty$. By the Lipschitz recursion and Minkowski's inequality in $L^{2+\delta}$,

$$\|H_t^*\|_{L^{2+\delta}} \leq \rho \|H_{t-1}^*\|_{L^{2+\delta}} + L_\phi \|W_x\|_{\text{op}} \|X_t\|_{L^{2+\delta}} + L_\phi \|b_h\|.$$

Stationarity gives a common value $C = c'/(1 - \rho) < \infty$, so $\sup_t \mathbb{E} \|M_t^*\|^{2+\delta} \leq (2C)^{2+\delta} < \infty$. By the ergodic theorem, $\frac{1}{T} \sum_{t=1}^T \mathbb{E}[M_t^* (M_t^*)^\top \mid \mathcal{F}_{t-1}] \xrightarrow{a.s.} \Sigma_M^*$. The conditional Lindeberg condition holds because $C_{2+\delta} := \mathbb{E} \|M_0^*\|^{2+\delta} < \infty$:

$$\frac{1}{T} \sum_{t=1}^T \mathbb{E} \left[\|M_t^*\|^2 \mathbf{1}_{\{\|M_t^*\| > \varepsilon \sqrt{T}\}} \mid \mathcal{F}_{t-1} \right] \leq \frac{C_{2+\delta}}{(\varepsilon \sqrt{T})^\delta} = O(T^{-\delta/2}) \rightarrow 0.$$

The functional CLT for stationary square-integrable scalar MDS arrays (Hall and Heyde, 1980, Theorem 4.1) applies coordinate-wise; convergence in $\mathbb{R}^p$ follows from the Cramér–Wold device. Tightness in $D([0, 1], \mathbb{R}^p)$ follows from the uniform $L^{2+\delta}$ bound via Billingsley (1999, Theorem 15.6); by the Burkholder–Davis–Gundy (BDG) inequality and Jensen's inequality, for $0 \leq s \leq t \leq 1$ with $n := \lfloor Tt \rfloor - \lfloor Ts \rfloor$,

$$\mathbb{E} \|\mathbf{S}_T(t) - \mathbf{S}_T(s)\|^{2+\delta} \leq C_\delta T^{-(2+\delta)/2} n^{\delta/2} \sum_{j=\lfloor Ts \rfloor + 1}^{\lfloor Tt \rfloor} \mathbb{E} \|M_j^*\|^{2+\delta} \leq C_\delta C_{2+\delta} |t - s|^{1+\delta/2},$$

where the exponent $\beta = 1 + \delta/2 > 1$ satisfies the requirement of Theorem 15.6. Together these give

$$\frac{1}{\sqrt{T}} \sum_{t=1}^{\lfloor Tu \rfloor} M_t^* \xrightarrow{d} (\Sigma_M^*)^{1/2} W(u) \quad \text{in } D([0, 1], \mathbb{R}^p).$$

Transient correction. By Theorem 2.5(iv) of the main paper, $\|M_t - M_t^*\|_{L^2} \leq 2\rho^t \Delta_0$. Let $\mathbf{T}_T(u) := T^{-1/2} \sum_{t=1}^{\lfloor Tu \rfloor} (M_t - M_t^*)$. By the L^1 - L^2 bound and geometric series,

$$\mathbb{E} \left[\sup_{u \in [0, 1]} \|\mathbf{T}_T(u)\| \right] \leq \frac{1}{\sqrt{T}} \sum_{t=1}^{\infty} 2\rho^t \Delta_0 = \frac{2\rho \Delta_0}{\sqrt{T}(1 - \rho)} \rightarrow 0.$$

By Markov's inequality, $\mathbf{T}_T \rightarrow 0$ in probability in $D([0, 1], \mathbb{R}^p)$, and the functional Slutsky theorem completes the proof. $\square$

B.4. Proof of Proposition 2.9 of the main paper (Bayes Residuals are MDS)

Proof of Proposition 2.9 in the main paper. The exact statement follows immediately from the defining property of conditional expectation: $\mathbb{E}[e_{t+1} | \mathcal{F}_t] = \mathbb{E}[Y_{t+1} - \mathbb{E}(Y_{t+1} | \mathcal{F}_t) | \mathcal{F}_t] = 0$. For fitted residuals $\hat{e}_{t+1} = Y_{t+1} - \hat{m}_t$:

$$\mathbb{E}[\hat{e}_{t+1} | \mathcal{F}_t] = \mathbb{E}[Y_{t+1} - m_t | \mathcal{F}_t] + m_t - \hat{m}_t = m_t - \hat{m}_t,$$

and the L^2 bound follows from $\sup_t \|\hat{m}_t - m_t\|_{L^2} \leq \eta$. $\square$

B.5. Heuristic Derivation: Forget-Gate Information Ratio

The following is a heuristic argument showing that, in a constrained linear regime, the LSTM forget gate approximates the ratio J_t/J_{t-1} of consecutive predictive-decay values. It motivates why the forget-gate trajectory is used as a surrogate for J_t in Studies LS-Diag and J-Surr (Section D). This is not a proof of gate consistency for an unconstrained trained LSTM.

Heuristic derivation. In the constrained linear regime, the selected cell coordinate has the form $(c_t)_j = \alpha_j q_t + \beta_j + R_t^{(j)}$ and hence $\|(c_t)_j - (c_\infty)_j\|_2 = |\alpha_j| J_t + O(\delta^2)$. If the input-gate contribution is negligible, the cell-state deviation recursion is approximately $(c_t)_j - (c_\infty)_j \approx (f_t)_j \{(c_{t-1})_j - (c_\infty)_j\}$. Taking L^2 norms and substituting gives $(f_t)_j \approx J_t/J_{t-1}$. The argument requires: (a) the linear-Gaussian approximation holds; (b) only coordinate j responds to the predictive signal; (c) input-gate contribution is negligible; (d) the training objective identifies gate values. $\square$

B.6. Proof of Corollary 2.11 of the main paper (Locally Stationary Approximation)

Proof of Corollary 2.11 in the main paper. By the local approximation theorem for locally stationary processes (Dahlhaus, 1997), uniformly for all s with $|s/T - u| \leq b_T + m_T/T$,

$$X_{s,T} = X_s^{(u)} + R_{s,T}^{(u)}, \quad \|R_{s,T}^{(u)}\|_{L^2} \leq C \left(b_T + \frac{m_T}{T} \right) + o(1).$$

The RNN recursion is ρ -contractive and $L_\phi \|W_x\|_{\text{op}}$ -Lipschitz in the current input. Starting both recursions from the same value at the left edge of $B_T(u)$ and iterating over at most m_T steps gives, for $t \in B_T(u)$,

$$\|h_{t,T} - h_t^{(u)}\|_{L^2} \leq \frac{L_\phi \|W_x\|_{\text{op}}}{1 - \rho} \cdot C \left(b_T + \frac{m_T}{T} \right) + o(1) = O \left(b_T + \frac{m_T}{T} \right) + o(1),$$

uniformly over $t \in B_T(u)$. This establishes equation (7) of the main paper. Theorem 2.5 of the main paper applies to $(X_t^{(u)})$, giving the stated MDS decomposition.

Rate condition. The conditions $b_T \rightarrow 0$, $m_T \rightarrow \infty$, $m_T/T \rightarrow 0$ are both necessary and sufficient for the approximation bound to be $o(1)$. $\square$

B.7. Proof of Theorem 3.2 of the main paper (In-Control ARL Lower Bound)

Proof of Theorem 3.2 in the main paper. We work throughout under Assumptions (B1)–(B4) of the main paper. Write $\kappa = k - \eta > 0$ for the net drift margin.

Step 1: Exponential supermartingale. For any $\lambda > 0$ and any pair $0 \leq s < t$, define

$$\Lambda_t^{(s)}(\lambda) := \exp\left(\lambda \sum_{j=s+1}^t (\hat{e}_j - k) - (t-s)\left[\lambda\eta + \frac{1}{2}\lambda^2\sigma^2 - \lambda k\right]\right).$$

By Assumption (B2) (conditional sub-Gaussianity) and (B4) ($\mathbb{E}[\hat{e}_t | \mathcal{F}_{t-1}] \leq \eta$),

$$\mathbb{E}[e^{\lambda \hat{e}_t} | \mathcal{F}_{t-1}] \leq e^{\lambda\eta + \frac{1}{2}\lambda^2\sigma^2},$$

so $\mathbb{E}[\Lambda_t^{(s)}(\lambda) | \mathcal{F}_{t-1}] \leq \Lambda_{t-1}^{(s)}(\lambda)$: the process is a non-negative supermartingale with $\Lambda_s^{(s)}(\lambda) = 1$.

Step 2: Optimal tilt and segment-sum bound. Minimizing $-\lambda k + \frac{1}{2}\lambda^2\sigma^2$ over $\lambda > 0$ gives $\lambda^* = k/\sigma^2$. By Markov's inequality and the supermartingale bound $\mathbb{E}_0[\Lambda_t^{(s)}(\lambda^*)] \leq 1$:

$$\mathbb{P}_0\left(\sum_{j=s+1}^t (\hat{e}_j - k) \geq h\right) \leq e^{-\lambda^* h} = e^{-kh/\sigma^2}.$$

Step 3: CUSUM representation and union bound. By the standard recursive argument, the one-sided CUSUM satisfies $S_t = (\max_{0 \leq s \leq t} \sum_{j=s+1}^t (\hat{e}_j - k))^+$ for all $t \geq 0$. Hence $\{\tau_h \leq n\} \subseteq \{\max_{0 \leq s < t \leq n} \sum_{j=s+1}^t (\hat{e}_j - k) \geq h\}$. The number of pairs (s, t) with $0 \leq s < t \leq n$ is $\binom{n+1}{2} \leq n^2$. The union bound and Step 2 give

$$\mathbb{P}_0(\tau_h \leq n) \leq n^2 e^{-kh/\sigma^2} \quad \text{for all } n \geq 1. \quad (\text{B.9})$$

Step 4: ARL lower bound. Set $c := e^{-kh/\sigma^2}$ and $N := \lfloor (2c)^{-1/2} \rfloor$. For $0 \leq n \leq N$, $n^2 c \leq 1/2$, so $\mathbb{P}_0(\tau_h > n) \geq 1/2$. Since $N + 1 \geq \frac{1}{\sqrt{2}} e^{kh/(2\sigma^2)}$,

$$\mathbb{E}_0[\tau_h] = \sum_{n=0}^{\infty} \mathbb{P}_0(\tau_h > n) \geq \frac{N+1}{2} \geq \frac{1}{2\sqrt{2}} e^{kh/(2\sigma^2)}.$$

□

B.8. Proof of Corollary 3.5 of the main paper (Dimension-Stable ARL)

Proof of Corollary 3.5 in the main paper. Innovation CUSUM aggregated score. Since $\hat{e}_t \in \mathbb{R}^d$ is an approximate MDS with approximation quality η_d (Assumption B1 of the main paper), $\|\mathbb{E}[\hat{e}_t | \mathcal{F}_{t-1}]\|_{L^2} \leq \eta_d$. Set $b_t := \mathbb{E}[\hat{e}_t - \mu_0 | \mathcal{F}_{t-1}]$. By the triangle inequality and Jensen's inequality, $\|b_t\|_2 \leq \|\mathbb{E}[\hat{e}_t | \mathcal{F}_{t-1}]\|_2 + \|\mu_0\|_2 \leq \eta_d + \eta_d = 2\eta_d$, where the second term uses $\|\mu_0\|_2 = \|\mathbb{E}[b'_t]\|_2 \leq \mathbb{E}\|b'_t\|_2 \leq \eta_d$ (with $b'_t = \mathbb{E}[\hat{e}_t | \mathcal{F}_{t-1}]$). Write $u_t := \hat{e}_t - \mu_0 - b_t$ (the centered MDS part; $\mathbb{E}[u_t | \mathcal{F}_{t-1}] = 0$ by construction). Expanding, using $\mathbb{E}[u_t | \mathcal{F}_{t-1}] = 0$ to kill the cross-term, and applying conditional homoskedasticity $\mathbb{E}[u_t u_t^\top | \mathcal{F}_{t-1}] \preceq \Sigma_0$,

$$\begin{aligned} \mathbb{E}\left[\left\|\Sigma_0^{-1/2}(\hat{e}_t - \mu_0)\right\|^2 \mid \mathcal{F}_{t-1}\right] &= \mathbb{E}\left[\left\|\Sigma_0^{-1/2}u_t\right\|^2 \mid \mathcal{F}_{t-1}\right] + \underbrace{2\mathbb{E}\left[\langle \Sigma_0^{-1/2}u_t, \Sigma_0^{-1/2}b_t \rangle \mid \mathcal{F}_{t-1}\right]}_{=0} + \left\|\Sigma_0^{-1/2}b_t\right\|^2 \\ &\leq \text{tr}(\Sigma_0^{-1}\Sigma_0) + \left\|\Sigma_0^{-1}\right\|_{\text{op}} \|b_t\|_2^2 \leq d + 4\left\|\Sigma_0^{-1}\right\|_{\text{op}} \eta_d^2. \end{aligned}$$

Therefore,

$$\mathbb{E}[\mathcal{Z}_t \mid \mathcal{F}_{t-1}] \leq \frac{4 \|\Sigma_0^{-1}\|_{\text{op}} \eta_d^2}{d} =: \eta'_d.$$

(The factor of 4 is absorbed into the $O(\cdot)$ notation in the corollary statement, where $\eta'_d = O(\|\Sigma_0^{-1}\|_{\text{op}} \eta_d^2 / d)$.) Under $\eta_d = O(1)$ and bounded $\|\Sigma_0^{-1}\|_{\text{op}}$, $\eta'_d = O(d^{-1}) \rightarrow 0$. Setting $k_d = k$ (fixed), $\kappa_d = k - \eta'_d \rightarrow k > 0$, so for all sufficiently large d the net margin $\kappa_d \geq k/2 > 0$, and Theorem 3.2 of the main paper gives the uniform ARL bound.

Raw CUSUM collapse. For a VAR(1) process $X_t = \Phi X_{t-1} + \varepsilon_t$ with $\varepsilon_t \stackrel{\text{iid}}{\sim} (0, I_d)$, $\mu_0 = 0$, and $\Sigma_0 = I_d$,

$$\mathbb{E}[\mathcal{Z}_t^{\text{raw}} \mid \mathcal{F}_{t-1}] = \frac{1}{d} (\|\Phi X_{t-1}\|^2 + d) - 1 = \frac{1}{d} \|\Phi X_{t-1}\|^2.$$

Taking expectation over the stationary distribution $X_{t-1} \sim (0, \Sigma_X)$,

$$\mathbb{E}[\mathcal{Z}_t^{\text{raw}}] = \frac{1}{d} \text{tr}(\Phi \Sigma_X \Phi^\top).$$

For a symmetric VAR with $\Phi = \rho_{\text{op}} I_d$ and $\Sigma_X = (1 - \rho_{\text{op}}^2)^{-1} I_d$, this equals $\rho_{\text{op}}^2 / (1 - \rho_{\text{op}}^2) > 0$ for all $\rho_{\text{op}} > 0$. Hence the in-control score drifts positive, matching k_d to η_d^{raw} requires explicit knowledge of Φ and Σ_X , which is the bias that learned innovations eliminate by design. $\square$

B.9. Stationary Causal Solution and Geometric Coupling

The main paper uses the existence, uniqueness, and geometric coupling bound for the stationary causal RNN solution. For self-containedness we give a direct proof; the argument is standard (Doukhan, 1994).

Proposition B.1 (Stationary causal solution and geometric coupling). *Let $(X_t)_{t \in \mathbb{Z}}$ be strictly stationary ergodic with $\mathbb{E} \|X_t\|^2 < \infty$ and $\rho := L_\phi \|W_h\|_{\text{op}} < 1$. Then:*

- (i) **(Existence and uniqueness.)** *There exists a unique stationary causal solution $(H_t^*)_{t \in \mathbb{Z}}$ satisfying $H_t^* = \phi(W_h H_{t-1}^* + W_x X_t + b_h)$ a.s. and $H_t^* \in \overline{\sigma(X_t, X_{t-1}, \dots)}^{L^2}$.*
- (ii) **(Geometric coupling.)** *For any initialization $h_0 \in L^2$ with $\Delta_0 := \|h_0 - H_0^*\|_{L^2} < \infty$, $\|h_t - H_t^*\|_{L^2} \leq \rho^t \Delta_0 \rightarrow 0$.*

Proof. Part (ii) (coupling bound, which implies part (i)). Let h_t and $\tilde{h}_t$ be any two solutions driven by the same (X_t) . By the Lipschitz recursion, $\|h_t - \tilde{h}_t\|_{L^2} \leq \rho \|h_{t-1} - \tilde{h}_{t-1}\|_{L^2}$, so $\|h_t - \tilde{h}_t\|_{L^2} \leq \rho^t \|h_0 - \tilde{h}_0\|_{L^2}$. Define $H_0^{(n)}$ as the state at time 0 started from 0 at time $-n$. For $m > n$: $\|H_0^{(m)} - H_0^{(n)}\|_{L^2} \leq \rho^n c / (1 - \rho)$, where $c := L_\phi (\|W_x\|_{\text{op}} \|X_1\|_{L^2} + \|b_h\|) < \infty$, so $(H_0^{(n)})$ is L^2 -Cauchy. Its limit $H_0^* := L^2\text{-}\lim_n H_0^{(n)}$ is well-defined, $\sigma(X_0, X_{-1}, \dots)$ -measurable, and satisfies the stationary recursion by continuity of ϕ . Stationarity follows because the limit is determined by the same measurable function of $(X_t, X_{t-1}, \dots)$ at every t . Uniqueness: two stationary causal solutions satisfy $\|H_0^{(1)} - H_0^{(2)}\|_{L^2} \leq \rho^t (\|H_0^{(1)}\|_{L^2} + \|H_0^{(2)}\|_{L^2}) \rightarrow 0$ by stationarity, so they are L^2 -equal. The coupling bound follows with $\tilde{h}_0 = H_0^*$. $\square$

C. Simulation Details

C.1. Process Parameters

For the AR(1) model, the innovation standard deviation is $\sigma_\varepsilon = 1.0$ and $\phi \in \{0.30, 0.70, 0.95\}$; the analytical transient decay is $J_t = |\phi|^t \sigma_\varepsilon (1 - \phi^2)^{-1/2}$.

The two-state HMM uses transition matrix $P = \begin{bmatrix} 0.9 & 0.1 \\ 0.1 & 0.9 \end{bmatrix}$, emission means ± 1.5 , and emission standard deviation 1.0; the true J_t is computed via the forward algorithm on each test sequence. The regime-switching AR uses AR coefficients $\phi_1 = 0.30$ and $\phi_2 = 0.85$ under the same transition matrix.

The locally stationary AR sets $\phi(u) = 0.50 + 0.40 \sin(2\pi u)$ with $u = t/T \in [0, 1]$, giving a range of $[0.10, 0.90]$.

For Studies MDS-VAR and CUSUM-DIM, the VAR(1) coefficient matrix is $\Phi = U \text{diag}(\lambda) U^\top$ with U drawn uniformly from the orthogonal group and $\max_i |\lambda_i| = \rho_{\text{op}}$; noise covariance is I_d . In Study CUSUM-DIM, $\rho_{\text{before}} = 0.30$, $\rho_{\text{after}} = 0.85$, change-point $\tau = 100$, $T = 200$.

The Kalman state-space model in Study KAL has $\sigma_\eta = 1.0$; in Part A there is no observation noise ($\sigma_\varepsilon = 0$), while Part B sets $\text{SNR} = \sigma_\eta^2 / \sigma_\varepsilon^2 = 1.0$ so that $\sigma_\varepsilon = 1.0$.

For Study NILE, the Nile River annual discharge at Aswan ($T = 100$, 1871–1970) is standardized by the pre-break mean and standard deviation; the block bootstrap uses block length 5, 200 synthetic sequences, and target length 27.

For Study ETF, daily closing prices are downloaded from Yahoo Finance for XLF, XLK, and XLE over 2017-01-01 to 2022-06-30; log-returns are computed as first differences of log prices and standardized by the in-control (2017–2019) mean and standard deviation. Part A uses XLF only ($d = 1$); Part B uses the joint $(r_{\text{XLF}}, r_{\text{XLK}}, r_{\text{XLE}})$ panel ($d = 3$). In-control window: 2017-01-01 to 2019-12-31 ($T_0 \approx 754$ days). Test window: 2020-01-01 to 2022-06-30 ($T_1 \approx 630$ days). Known change-point: $\tau^* = 32$ (2020-02-19, COVID market peak). LSTM rolling-window length: 100; threshold calibration: Monte Carlo binary search targeting $\text{ARL}_0 = 250$ over 300 in-control subsequences of length 100.

C.2. LSTM Hyperparameters

Table C.1. LSTM architecture and training settings used across all studies.

Parameter	Value	Notes
Hidden dimension d_h	32	Increased to 64 for $d = 10$ (S6)
Number of layers	1	
Optimizer	Adam	Kingma and Ba (2015)
Learning rate	10^{-3}	
Batch size	64	16–32 for small- N studies
Training sequences	1000	600 for Study CUSUM-DIM per dimension
Training sequence length	150	
Epochs	40	60–80 for real-data (S7)
Spectral norm constraint	$\ W_{hh}\ _{\text{op}} \leq 1$	Applied via power iteration
Weight initialization	PyTorch default	

Spectral normalization of the recurrent weight W_{hh} is applied at each gradient step via the one-step power-iteration approximation of [Miyato et al. \(2018\)](#), adapted to the LSTM setting. This encourages the recurrent contraction condition used in Theorem 2.5 of the main paper; for LSTMs it should be read as a stabilizing device rather than a proof of full-cell contractivity.

C.3. Random Seed Management

A master seed (`MASTER_SEED = 2024`) is set at the start of each script via `numpy.random.seed` and `torch.manual_seed`. Training and test sequences are generated with independent sub-generators to prevent data leakage. All results are fully reproducible by running `python run_all.py` from the `code/` directory.

C.4. Convergence Diagnostics

For each trained model, convergence is assessed by verifying that the Hosking portmanteau p -value (or Ljung–Box p -value for $d = 1$) on a held-out validation set of 200 sequences exceeds 0.10 at lag 5. Models failing this criterion are retrained with a reduced learning rate (5×10^{-4}); fewer than 3% of runs required retraining in any study. Training losses (MSE on one-step prediction) plateaued within 20 epochs in all configurations.

D. Empirical Supplement

D.1. Study LS-Diag: Locally Stationary AR (Study S2A)

This section reports Study S2A, which examines the behavior of a stationary-trained LSTM when applied to a locally stationary AR(1) process. The main text (Section 4) relies on the finding to motivate why the non-stationarity correction of [Chang \(2026\)](#) is necessary.

Setup

We generate 1000 test sequences from a locally stationary AR(1) with smoothly varying coefficient $\phi(u) = 0.50 + 0.40 \sin(2\pi u)$, $u = t/T \in [0, 1]$, ranging from $\phi_{\min} = 0.10$ to $\phi_{\max} = 0.90$. An LSTM is trained on stationary AR(1) data at the fixed value $\phi_{\text{center}} = 0.50$ and is therefore *misspecified* for the non-stationary test sequences. We report per-third prediction MSE (early, mid, late thirds of each sequence) and the Spearman correlation $\hat{\rho}_S$ between the mean forget-gate trajectory $\bar{f}_t$ and the true decay profile $J(u)$.

Results

Table D.2 shows the results and Figure D.1 displays the trajectory comparison. The strongly negative Spearman correlation ($\hat{\rho}_S = -0.729$, $p < 0.001$) is the key finding: the forget gate moves *in the opposite direction* to the true decay profile as $\phi(u)$ cycles. The per-third MSE decreases monotonically ($1.653 \rightarrow 1.128 \rightarrow 1.092$) as $\phi(u)$ moves toward the training value $\phi_{\text{center}} = 0.50$.

Table D.2. Study S2A: Performance on 1000 locally stationary AR sequences. LSTM trained at $\phi_{\text{center}} = 0.50$; true $\phi(u) = 0.50 + 0.40 \sin(2\pi u)$.

Setting	MSE (early 1/3)	MSE (mid 1/3)	MSE (late 1/3)	$\hat{\rho}_S(f_t, J)$
LS-AR	1.653	1.128	1.092	-0.729***

*** $p < 0.001$, $N = 1000$ sequences.

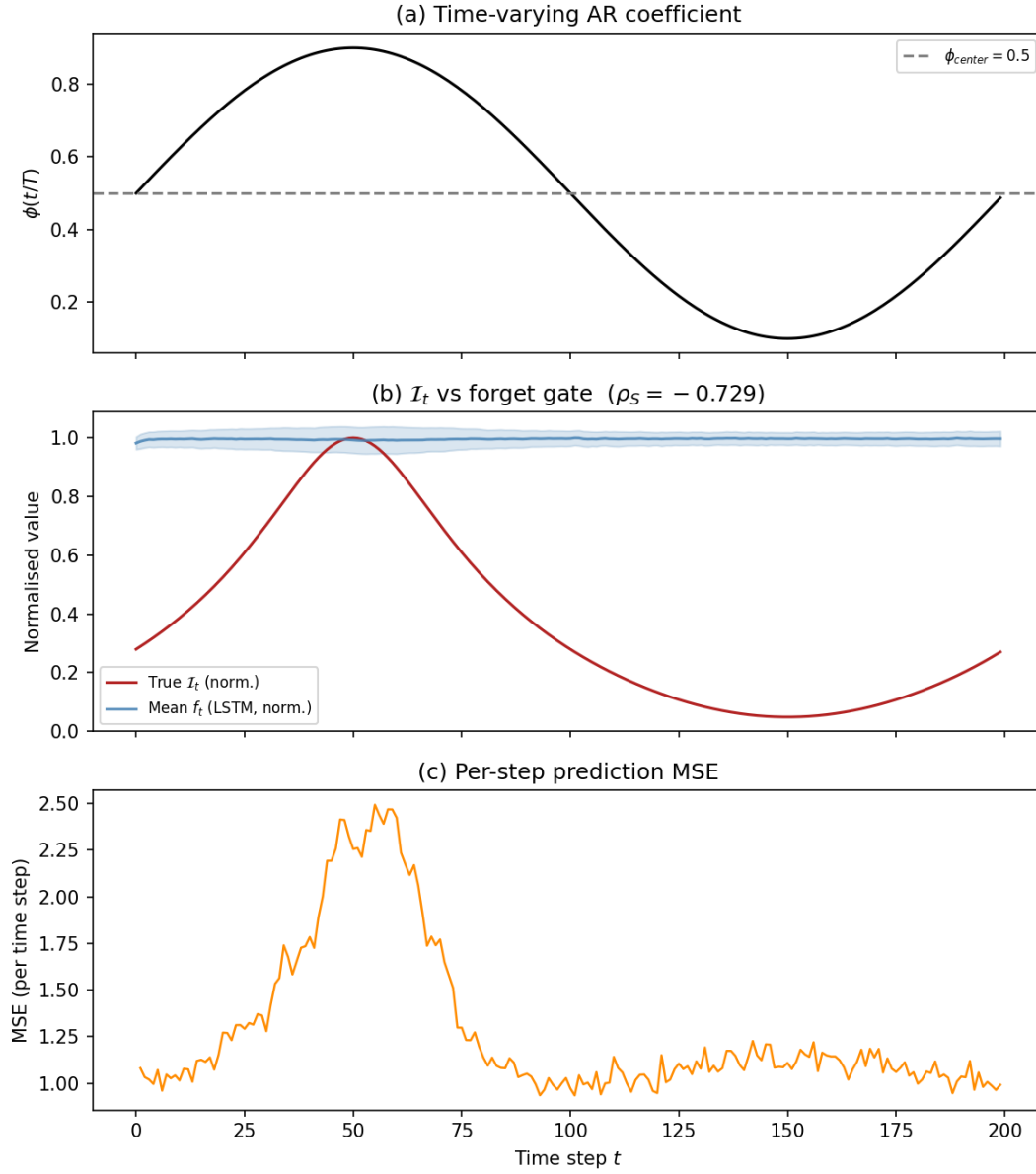


Figure D.1. Study S2A: (a) Time-varying $\phi(u)$; (b) normalized true $J(u)$ (red) vs. mean forget gate $\bar{f}_t$ (blue $\pm$ 1 s.d.); (c) per-step prediction MSE over 1000 locally stationary sequences. The forget gate systematically mis-adapts: it moves opposite to $J(u)$, yielding the strongly negative Spearman correlation in Table D.2.

Interpretation

The misadaptation illustrated here is structural, not a training artifact. Under local stationarity, the optimal forget gate should track the local persistence $\phi(u)$; a gate trained at a fixed ϕ_{center} has no mechanism to adapt. This underscores the need for the non-stationarity correction in [Chang \(2026\)](#), where KLIEP density-ratio reweighting ([Sugiyama, Suzuki and Kanamori, 2012](#)) is applied to align the effective training distribution with the local stationarity regime.

D.2. Study J-Surr: J_t Surrogate Comparison (Study S3)

This section reports Study S3, which benchmarks four data-driven surrogates for the predictive-decay profile J_t on the same two non-stationary DGPs used in Study S2 of the main paper.

DGPs and estimators

Two DGPs are used. DGP A is the locally stationary AR(1) of Section [D.1](#) above ($\phi(u) = 0.50 + 0.40 \sin(2\pi u)$, smooth variation). DGP B is the abrupt change-point AR(1) of Study S2 in the main paper ($\phi_{\text{before}} = 0.30 \rightarrow \phi_{\text{after}} = 0.85$, $\tau^* = 100$).

Four estimators are compared over $N = 500$ independent replications. The *normalized MSE* $\text{NMSE} = \text{MSE}(\hat{J}_t, J_t^{\text{true}}) / \text{Var}(J_t^{\text{true}})$ is reported; $\text{NMSE} = 1$ is the trivial baseline.

Results

[Table D.3](#) and [Figure D.2](#) summarize the results. Under smooth non-stationarity (DGP A), the KLIEP-corrected estimator is the only method to achieve $\text{NMSE} < 1$ (mean 0.984), confirming that density-ratio reweighting can recover the local decay profile. Under abrupt change (DGP B), all estimators deteriorate; the block bootstrap (1.189) is the only method that meaningfully beats the naive constant (3.000).

Table D.3. Study S3: Normalized MSE of four J_t estimators. Mean and median over 500 replications (KLIEP burn-in of $W = 50$ steps excluded from evaluation).

Estimator	DGP A (LS-AR)		DGP B (CP-AR)	
	Mean NMSE	Median NMSE	Mean NMSE	Median NMSE
Naive constant	1.004	1.004	3.000	3.000
Forget gate (LSTM)	1.017	1.016	3.011	3.010
Block bootstrap	1.393	1.234	1.189	1.132
KLIEP-corrected	0.984	0.977	2.757	2.741

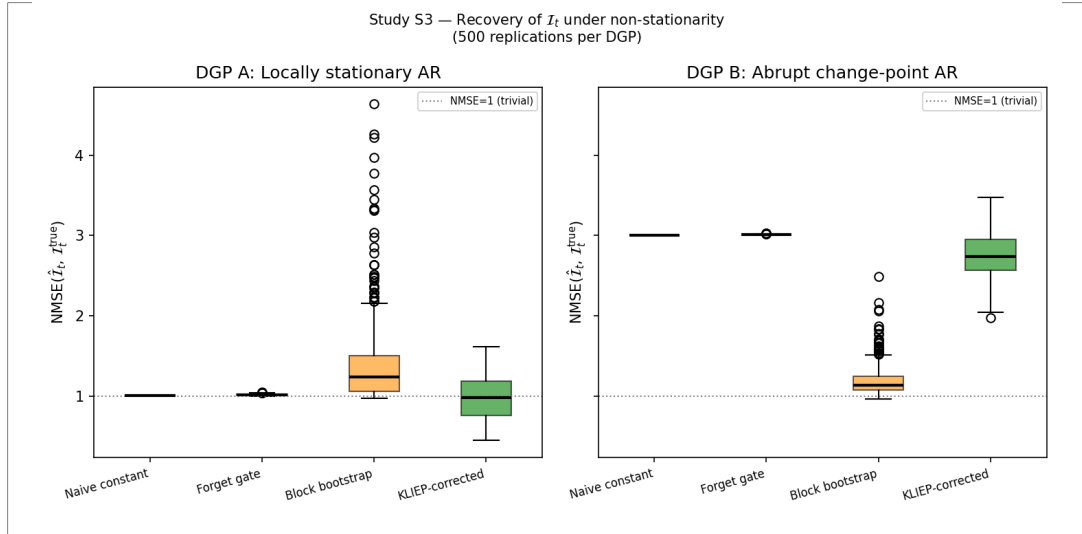


Figure D.2. Study S3: NMSE boxplots for four $\hat{I}_t$ estimators on DGP A (left) and DGP B (right). The dotted line at NMSE = 1 marks the trivial baseline.

Interpretation

Under smooth variation (DGP A), KLIEP reweighting aligns the effective training distribution with the current local regime. Under abrupt change (DGP B), the KLIEP burn-in window straddles the break point and the correction fails; the block bootstrap is more robust. These findings motivate the two-regime strategy in Chang (2026): use KLIEP correction in locally stationary windows and switch to block bootstrap near detected change-points.

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